Perceived Mental Workload Under Differing Reinforcement Structures in Game-Based Assessment

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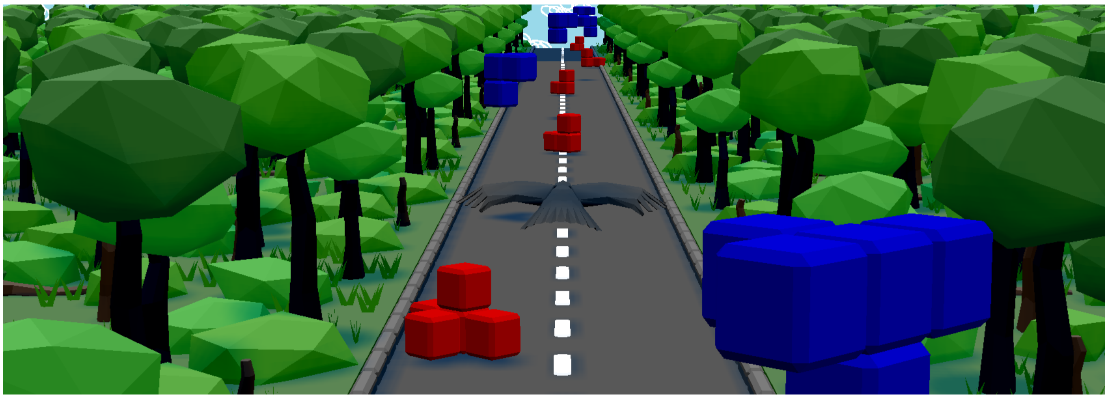

Figure 1: Screenshot of GBA Task.

Abstract

Game-based assessments are becoming an increasingly popular format in education, but little is known about how the reinforcement structures inherent to them influence the perceived mental workload of users. The present study used a between-subjects experimental design to examine differences in perceived mental workload when manipulating reinforcement structure in the same in-house built game-based assessment. No effect was found via primary analysis (Kruskal-Wallis test), although exploratory analyses found evidence of an effect in that those given in-game rewards perceived that they performed better than those who received in-game punishments or both rewards and punishments. Recommendations for future research are given including sampling power, ergonomic analysis and covariance validation of the game-based assessment used, and other measures of subjective experience. This outcome helps to lay groundwork for methodological considerations in future research.

1 Introduction

1.1 Game Based Assessment

Digital games are a pervasive phenomenon in contemporary society. In recent decades, efforts to exploit this entertaining and interactive format for education and assessment purposes have increased drastically [5]. Digital games' ability to motivate and generate curiosity enables the optimization of performance in the educational environment [21]. Such presentations of learning and assessment material may offset some of the aversiveness inherent to mental effort in education [4]. Nonetheless and unsurprisingly, practical implementational barriers in this relatively new field persist [13] [21]. The magnitude in range, quantity, and quality of data which can be collected from game-based assessments (GBA) offers the potential for a deeper understanding of how people learn, and a more holistic assessment of test-takers' knowledge, understanding, or skills; beyond what is possible with traditional methods [8]. GBA differs from gamified assessment in that the former operates through the fabric of gameplay, and the latter is a (re)design technique of traditional assessment media [31]. Real-time data logging of decisions, behaviour, and meta-data are examples of this deeper data collection, as well as the ability to seamlessly analyse logged data for guiding the provision of further learning materials among other outcome-related decisions [26]. Furthermore, GBAs allow for a flexible format which can be tailored to the context of the subject, offering a more contextually relevant experience to test-takers both in subject-matter and presentation [8]. Extending from this, another edge of GBAs lies in their ability to assess modern competencies which traditional assessments fail to capture [27]; this is critical to assessment validity, which concerns how well the relevant phenomena are captured [7]. While GBA opens these potentially fruitful avenues, most examples still adhere to direct measures of learning (e.g. recall or accuracy), and those which are less direct remain underexamined [9]. Naturally, this is concerning for the construct validity of extracted data and how it is interpreted.

1.2 Reinforcement and Affect in Learning and Assessment

Although operant conditioning theory was pioneered many decades ago [25], its insights remain relevant to the modern educational environment [14]. The theory is premised on the observation that behavior changes due to its associated consequences; thus, provision of a consequence for a certain behavior will either increase or decrease its likelihood of occurring again [10]. Reinforcement in this context refers to efforts to increase a behavior; this can be achieved through: positive reinforcement – the addition of a pleasant stimulus; or negative reinforcement – the removal of an unpleasant stimulus. Conversely, punishment in operant theory refers to efforts to decrease a behavior, again this can be: positive – the addition of an unpleasant stimulus; or negative – the removal of a pleasant stimulus [10]. Theories of learning and educational psychology have, of course, developed, but this simple insight remains intact and even necessary as part of modern practice [2][15]. The use of reinforcement and punishment in education is known to influence both academic achievement and behaviour; from an affective perspective, positive reinforcement most notably in increasing motivation, and positive punishment – while controlling undesired behaviour – risking creating stress and anxiety [1]. While research on reward and punishment in education tends toward performance and behaviour as its primary and direct subject of inspection, these affective regressands have their own impact on assessment outcomes. Stress, confidence, and degree of engagement are all significant predictors of scores on tests as well established and validated as Raven's Progressive Matrices [22].

1.3 In-Game Reinforcement

While a strict operational definition for 'game' is difficult to pin down, the vast majority to some degree entail predictable consequences to the player's actions. Possibly to a greater extent, due to the nature of deterministic code, this is true of digital games – any action taken by a player produces an outcome which is systemically coherent to the game's rules; any incorporated probability or randomness remains within defined parameters of possible outcomes. And as with education, reward and punishment are used extensively. While in this context, this is done to enrich experience and boost retention, the product resembles the constituent parts of operant theory [19]. Common rewards include progression, rankings, and cosmetics [24]; common punishments include cessation, ability reduction, and reward removal [16]. As in education, these appear similarly influential on user affect, with studies showing the addition of reward and punishment to an identical base game altering the perception of aspects such as 'challenge' and 'immersion' and even positive and negative affect directly [30]. Within this overall effect there are distinctions, punishment appears to result in more rapid performance increases than reward [29]; although as has been stated, this has affective tradeoffs [1]. While the purpose of GBA differs from that of games for entertainment, they nonetheless operate through the same medium, and therefore necessarily make use of the reward and punishment structures inherent to this medium – the affective implications of this are an area underexamined.

1.4 Perceived Mental Workload

As we can see, GBA test-takers do not operate in an affective vacuum. And importantly for GBA, not only the task, but the context in which it is performed is constitutive of self-assessed perceived mental workload [23]. So, when shifting between traditional and game-based formats, item difficulty is not an individually sufficient focus for maintaining assessment validity. The most prevalent and well-established measure of perceived mental workload is the National Aeronautics and Space Administration Task Load Index–6 Item (NASA-TLX) [11]. While a somewhat contentious debate, this measure is at least equivalently, if not a more, reliable and valid tool than similar measures of subjective workload (e.g. SWAT) [28]. And certain criticisms do exist, concerning factors such as its validity across applications so far removed from its origin and validity across different response media (e.g. [18]); but the NASA-TLX's use across multiple disciplines and versatility of application make it an attractive choice for this interdisciplinary study [11]. For these reasons, this study will make use of the NASA-TLX without any adaptation to retain its validity, sensitivity, reliability [11].

1.5 The Present Study

From what has been presented, we can see that reinforcement structure in GBA is rarely analysed. The rewards and punishments which evidently affect performance, motivation, affect, and other perceptions, likely also influence perceived mental workload. We pose that this research is necessary to develop a more complete picture of GBA and how users negotiate with them. This negotiation, as seen in previous examples has a complex interplay with outcome scores, which necessitates further investigation to ensure validity in GBA, making use of research at the intersection of education, psychology and game development.

Here, we focus on positive reinforcement and positive punishment – that is, the addition of positive and negative stimuli to GBA. For simplicity of coding and distinction, these are henceforth referred to as the 'positive' and 'negative' conditions of the study. The present study aims to assess the extent to which reinforcement structure modulates perceived mental workload in game-based assessments. Based on the available literature, it is hypothesized that H1: the addition of reinforcement to the game-based assessment will produce changes in perceived mental workload. Analyses beyond those which assess H1 are exploratory.

2 Methods

2.1 Design and Methodology

This study employs a between-subjects experimental design in order to examine the predictive relationship between reinforcement structure in GBAs and perceived mental workload. As the sample size for this study is small and acts as a proof-of-concept, conventional statistical analyses such as ANOVA and t-tests have been avoided, this, along with which analyses will be included is discussed in the 'Data Analysis' section below. This design is appropriate as we are examining potential changes in an outcome variable when isolated and intentional changes are made to a single aspect of a task. In doing this, the study aims to provide insight into an area of interest and importance, acknowledging that other methods may be required to better understand the mechanisms of this relationship – see 'Discussion'.

2.2 Participants

G*Power analysis (v3.1.9.7) indicates that for a medium effect size of f = 0.25 with an α error probability of 0.05, and detection power of 1-β = 0.8, a total sample size of at least 180 is required to individually assess the influence of each of the four reinforcement structures using conventional analyses; due to a lack of similar research, these input values follow the guidelines of conventional behavioral research [3]. As this was not viable for this short pilot study, the participants were drawn from a convenience sample of acquaintances of the primary author who are all enrolled at Auckland University of Technology (AUT) and the University of Auckland (UoA) and alternative analyses are used (see 'Data Analysis'). This selection was appropriate as this study serves as a pilot for potential future research. There were no necessary exclusion criteria within this sample. The final sample included 16 AUT and UoA students, assigning 4 to each condition. Participants were blinded to conditions.

2.3 The GBA Task

The game-based assessment used in this study was designed in-house. It was developed using Python 3.13.3 with the Ursina game engine (v8.1.1); assets for the game were created using Blender (4.5.2). This allowed total control over task parameters and real-time behavioral and performance data logging. As the focus of this research was perceived mental workload across different reinforcement structures, the construct, content, and criterion validity of assessment items were not of primary importance. However, face validity remained important in order to ensure the GBA felt to participants as though it was indeed a test. The task was designed to impose several cognitive demands so as to simulate moderate mental workload – this was necessary so that changes to reinforcement structure may have a discriminable effect on PMW. These cognitive demands included: inhibitory control, selective attention, cognitive flexibility in rule maintenance, mental rotation, and performing under time pressure.

To achieve this, the task entailed controlling the Y-axis (up and down) and X-axis (left and right) movement of an avatar in a fixed-speed continuous side-scrolling environment. Polycube objects of three sorts (i_1, i_2, i_3) were randomly selected and generated alternately on the ceiling (in blue) and floor (in red) at random 90° interval rotations about the Y-axis. The spatial intervals between objects were such that it is always possible to collect the next item. Prior to the game beginning, an instruction screen indicated that i_1 should be collected in red while avoiding all other red objects, and i_1 should be avoided in blue while collecting all other blue objects.

This format provided the foundation for the four games which made up each experimental condition. One condition was strictly time-based and added no reinforcement. The positive reinforcement condition added the release of coins from correctly intersected and correctly avoided items, along with an accumulating coin counter and positive sound effects [24]. The negative reinforcement condition added a health bar to the side of the screen which decreased with incorrect intersections and incorrectly missed items along with red flashes across the screen and negative buzzer sound effects [24]. The final condition added all aspects of both the positive and negative conditions (See figure 2 for example).

In all four games, after a visible three-minute timer had elapsed, the game ended, and participants were automatically directed to the NASA-TLX survey panel (see 'Perceived Mental Workload').

Figure 2: Reinforcement in the GBA (cropped)

2.4 Perceived Mental Workload

The measure used for perceived mental workload was the NASA-TLX survey retrieved from the NASA Human Systems Integration Division website [12]. This consists of six items answered on a 0-20 scale covering mental demand, physical demand, temporal demand, performance, effort, and frustration. Each scale extends from 0 – very low – to 20 – very high, with the exception of performance which extends from 0 – perfect – to 20 – failure. Participants responded to these items on an in-game slider screen (See Figure 3). Upon submission of responses, the data was exported to a local .csv file, where the responses to each item were recorded alongside an arbitrary identifier and the game condition relevant to the responses. Participants were then thanked for their participation and debriefed.

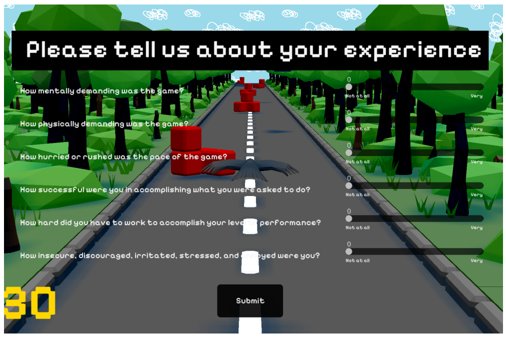

Figure 3: In-Game NASA-TLX survey (cropped)

2.5 Data Analysis

Data analysis was performed using R v4.5.1 through Rstudio v2025.05.1+513. T-tests and ANOVA have intentionally been omitted from this analysis due to the small sample size as these would likely yield insufficient statistical power, unreliable assumption checks for normality and homogeneity, and risk of type 1 and 2 errors [6]; as such the p-values from these analyses may be misleading [6]. The analysis begins with descriptive statistics and visualisations, then a Kruskal-Wallis test is used to determine if reinforcement structure significantly changes response distribution. If a significant difference is found, a Wilcoxon rank-sum test will be used where appropriate to locate the effect. Finally, condition median differences are estimated with 5000 bootstrap resamples to attain 95% confidence intervals for effect size. The Kruskal-Wallis test has been selected as it is non-parametric, rank-based, and is suitable for this study's four conditions [20]; Wilcoxon rank-sum is an appropriate post-hoc analysis as our conditions are independent, it utilises a pairwise methodology and again does not assume normality in the data; bootstrapped confidence intervals here are exploratory and descriptive of possible effect ranges rather than inferential due to the small n. This analysis plan will be performed for both total NASA-TLX score and per item.

2.6 Ethics

As this pilot study was only a proof-of-concept and involved only known acquaintances of the primary researcher, a formal ethics review was not required according to institutional guidance. All data recorded from participants was anonymous and stored on a password protected laptop. There was no risk or harm to participants beyond inconvenience of time taken. To uphold participant autonomy, participants were made aware of their anonymity prior to participation and were informed that they were able to withdraw from the study without penalty.

3 Results

3.1 Total NASA-TLX Responses

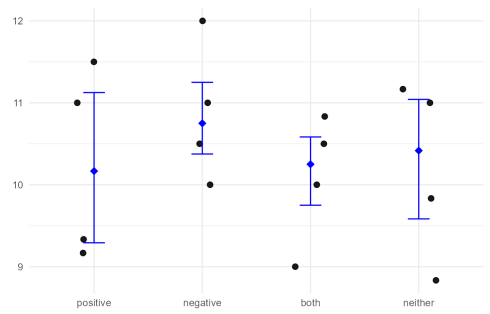

Figure 4: Total NASA-TLX Score by Reinforcement Structure

Raw response distribution data for total NASA-TLX survey is seen in Figure 4. A summary can be found in Table 1. Here it can be seen that the negative condition's score averages sit above those of the other conditions.

Table 1
Descriptive Statistics for Total NASA-TLX Responses.

	Median	Mean[SD]	Range
Positive	61	61.5[7.05]	55–69
Negative	64.5**	65.25[5.12]**	60–72
Both	61.5	60.5[4.80]	54-65
Neither	62.5	61.25[6.55]	53–67

Despite this, the Kruskal-Wallis test found no significant difference between reinforcement structures for the total TLX score p = 0.6843, and the post-hoc Wilcoxon rank-sum test was not run. The bootstrapped 95% CIs all contained 0, indicating a negligible difference between conditions (See Table 2).

Table 2
Bootstrapped Median Differences for Total NASA-TLX Responses with 95%CI

	Positive	Negative	Both
Negative	-0.583[-2.25–1.00]	–	–
Both	-0.083[-1.42–1.75]	0.5[-0.42–2.00]	–
Neither	-0.25[-1.83–2.17]	0.33[-0.83–2.17]	-0.17[-1.58–1.50]

3.2 Per Item Responses

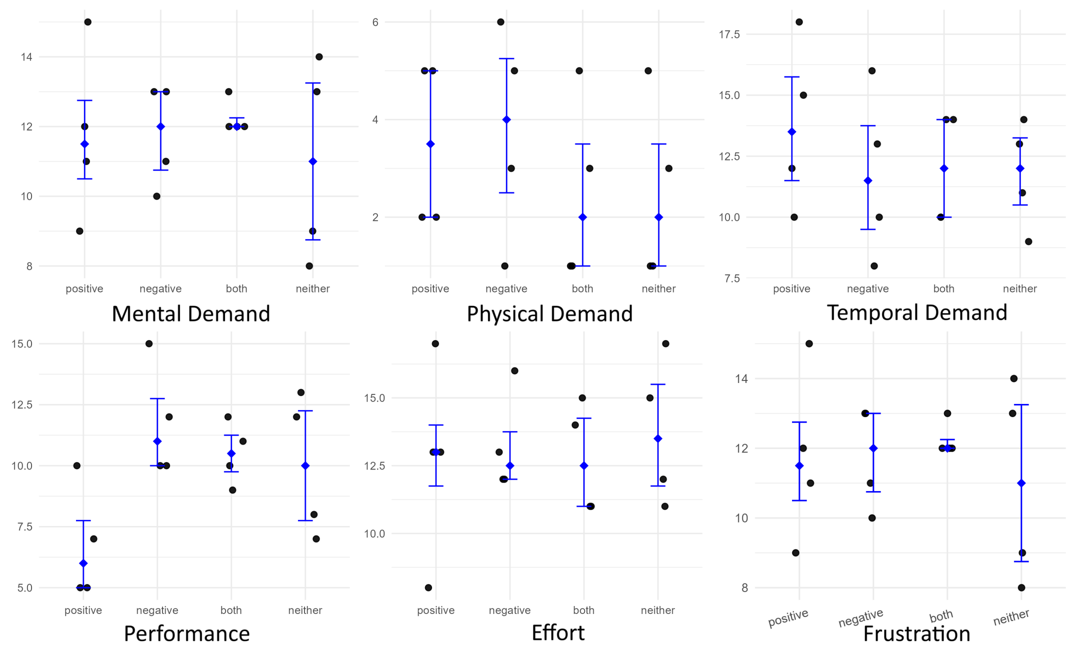

Figure 5: Item-wise NASA-TLX Score by Reinforcement Structure

Raw response distribution data for each NASA-TLX item are seen in Figure 5. Summaries are found in Table 3. Response averages per item are quite tightly grouped across conditions, with exceptions being found in the 'positive' condition responding with lower performance scores (indicating they perceived their performance as better), and in the 'both' condition responding with lower frustration.

The Kruskal-Wallis tests found no significant difference between reinforcement structures across any of the six items; all p-values were above 0.05. As such, no post-hoc Wilcoxon rank-sum tests were necessary nor run by the R script.

2 of the 36 bootstrapped CIs did not contain 0 (See Table 4) – these were both in the performance item – between the 'positive' and 'negative' conditions, and between the 'positive' and 'both' conditions. This indicates that there exists a nonzero difference between the median scores of these groups on this item. Nonetheless, as per the 'Data Analysis' section, this result should be considered exploratory due to the small n.

Table 3
Descriptive Statistics per NASA-TLX Item Response.

	Median	Mean[SD]	Range
Mental Demand			
Positive	11.5	11.75[2.5]	9–15
Negative	12	11.75[1.5]	10–13
Both	12	12.25[0.5]	12–13
Neither	11	11[2.94]	8-14
Physical Demand			
Positive	3.5	3.5[1.73]	2–5
Negative	4	3.75[2.22]	1–6
Both	2	2.5[1.91]	1–5
Neither	2	2.5[1.91] 1–5	
Temporal Demand			
Positive	13.5	13.75[3.5]	10–18
Negative	11.5	11.75[3.5]	8–16
Both	12	12[2.31]	10–14
Neither	12	11.75[2.22]	9–14
Performance			
Positive	6**	6.75[2.36]	5–10
Negative	11	11.75[2.36]	10–15
Both	10.5	10.5[1.29]	9–12
Neither	10	10[2.94]	7–13
Effort			
Positive	13	12.75[3.69	8–17
Negative	12.5	13.25[1.89]	12–16
Both	12.5	12.75[2.06]	11–15
Neither	13.5	13.75[2.75]	11–17
Frustration			
Positive	13	13[2.58]	10–16
Negative	13	13[3.37]	9–17
Both	9.5**	10.5[3.87]	7–16
Neither	12.5	12.25[2.06]	10–14

4 Discussion

This study aimed to examine change to perceived mental workload under differing reinforcement structures in GBA. Our study did not find sufficient evidence to support H1 via its primary analysis. We must therefore conclude that, within our parameters, reinforcement structure did not produce a change in perceived mental workload. Exploratory analyses found some evidence of an effect in the performance item; with those given positive reinforcement responding with greater perceived performance than those in the 'negative' and 'both' reinforcement conditions. This indicates that further research is necessary to explore an effect using other methods; especially with regard to how positive reinforcement influences perceived performance. As with any area in its research infancy, there exist many possible changes to the study design which may be fruitful. Foremost, as sample size was a limiting factor in this study, we recommend a sufficiently powered study; allowing for inferential testing using t-tests or ANOVA. Similarly, the sample drawn was certainly not random and may not have been a populationally representative one, making it difficult to rule out whether participants interacted with the game differently than the average test-taker. Accordingly, we recommend a more impartial sampling plan in future research. These would be necessary steps before drawing strong conclusions about the nature of the GBA used. Nonetheless, it is acknowledged that it is possible that the GBA used here, being in its first rendition, may not be optimised for this research. The cognitive demand it imposed may have been too great, the exposure time insufficient, or the game simply too unfamiliar. For this reason, we recommend that the GBA designed for this study undergo ergonomic analysis to its own end, using more qualitative methods of user feedback by formats such as short answer response or semi-structured interview. This could also be used to gather feedback on the reinforcement iterations used (coins and health), which while representative of common game reinforcers, may or may not have been sufficiently reinforcing in this instance. A further potential method to better assess the appropriateness of our GBA would be to track participant performance in-game alongside perceived mental workload. As

Table 4
Bootstrapped Median Differences Per NASA-TLX Item Responses with 95%CI

	Positive	Negative	Both
Mental Demand			
Negative	-0.5[-3–3.01]	–	–
Both	-0.5[-3–3]	0[-2–1]	–
Neither	0.5[-3.5–5]	1[-3–4.5]	1[-2–4]
Physical Demand			
Negative	-0.5[-3.5–3]	–	–
Both	1.5[-2–4]	2[-2–4.5]	–
Neither	1.5[-2–4]	2[-2–4.5]	0[-3–3]
Temporal Demand			
Negative	2[-3.5–8]	–	–
Both	1.5[-3–6.5]	-0.5[-5–4.5]	–
Neither	1.5[-2.5–6.5]	-0.5[-4.5–5]	0[-3.5–4]
Performance			
Negative	-5[-9– -1]**	–	–
Both	-4.5[-6.5– -0.5]**	0.5[-1.5–4.5]	–
Neither	-4[-7.5–1.01]	1[-2.5–6]	0.5[-3–4]
Effort			
Negative	0.5[-5–4.5]	–	–
Both	0.5[-4.5–4.5]	0[-2.5–3.5]	–
Neither	-0.5[-6.5–4]	-1[-4.5–3]	-1[-5–3]
Frustration			
Negative	0[-5–5]	–	–
Both	3.5[-3–7]	3.5[-4–7.51]	–
Neither	0.5[-3–4.5]	0.5[-3.5–5]	-3[-6–3.5]

per the 'Reinforcement and Affect in Learning and Assessment' and 'In-Game Reinforcement' sections, with change to reinforcement we should expect change in performance; thus, by observing covariance of perceived mental workload and performance, we could determine whether any absence of significant findings reflects structural issues in our GBA or is representative of the true effect [1][29] Equally, other measures of subjective experience exist. The NASA-TLX, while being a validated measure of perceived mental workload, does not address experiential factors such as test anxiety, motivation, or self-efficacy beliefs, which are known to influence test outcomes [22]. This is a distinct but equally important research direction and could make use of the Dundee Stress State Questionnaire (DSSQ) or similar [17]. While secondary to the validity issues raised, the flexible format of GBA allows for interesting research on changes to GBA presentation [8]. This could include changing the game environment, physics, perspective, avatar, or any other aspect of the game to look for changes in workload, anxiety, motivation, or performance. In conclusion, while GBA offers many benefits to learners and test-takers, they introduce many unknown variables which individually require assessment to determine how they fit into the greater portrait of a valid assessment. Although no effect was presently observed, this outcome provides valuable groundwork by underlining methodological intricacies and other considerations and directions which future GBA research must address. We hope that this study inspires efforts towards developing valid and useful game-based assessments.

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